Comptia N+ Assignment

[N+ Network Fundamental and building networks]

IP routing and routing protocols

Task:

❖ **1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.**

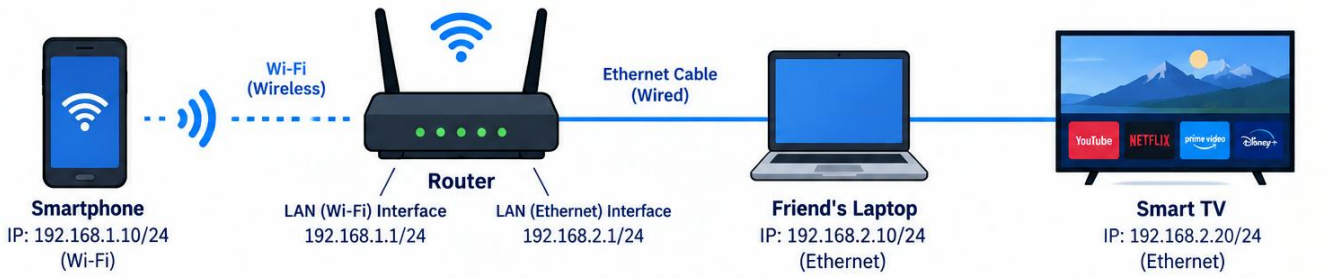
➢ **Aim**

- To draw a simple network diagram showing three devices connected through a router and label the IP addresses of each device and the router's interfaces.

➢ **Introduction**

- IP routing is the process of forwarding data packets from one network to another. A router connects different networks and uses IP addresses to determine where packets should be forwarded.
- Each device requires a unique IP address within its subnet. The router's interface IP address acts as the default gateway for devices on that network.

➢ Network Diagram



➢ 4. IP Address Assignment Table

Device / Interface	IP Address	Subnet Mask
Smartphone (Wi-Fi)	192.168.1.10	255.255.255.0
Router (Wi-Fi Interface)	192.168.1.1	255.255.255.0
Router (Ethernet Interface)	192.168.2.1	255.255.255.0
Friend's Laptop	192.168.2.10	255.255.255.0
Smart TV	192.168.2.20	255.255.255.0

➢ Explanation

1. The smartphone connects to the router through Wi-Fi using the IP address 192.168.1.10.
2. The friend's laptop and Smart TV connect to the router through wired Ethernet connections.

3. The router has two interfaces with different IP addresses: 192.168.1.1 and 192.168.2.1.
4. The router forwards packets between the two subnets according to its routing table.
5. The default gateway of each device is the IP address of the router interface on its own subnet.

- **Result**
- A network diagram containing three devices and a router was prepared. The IP addresses, subnet masks, and router interface addresses were assigned to demonstrate communication between two different subnets.

❖ **2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Use made-up IP addresses and show the routing table entries that would be checked at each hop.**

➢ **Aim**

- To understand how a packet travels from a laptop to a friend's computer in a different subnet by examining routing table entries and explaining the routing decisions at each hop.

➢ **Introduction**

- IP routing is the process of forwarding data packets from a source network to a destination network. Routers examine the destination IP address and use their routing tables to determine the next hop or outgoing interface.
- When two computers belong to different subnets, the source computer sends the packet to its default gateway. The router then forwards the packet toward the destination network.

➢ **Network Scenario**

➢ In this example, my laptop sends a packet to a friend's computer located in a different subnet. The packet passes through two routers before reaching the destination.

➢ **Source Laptop**

- IP Address: 192.168.1.10

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

➢ **Router R1**

- LAN Interface: 192.168.1.1
- Link to R2: 10.0.0.1

➢ **Router R2**

- Link to R1: 10.0.0.2
- Destination LAN Interface: 192.168.2.1

➢ **Friend's Computer**

- IP Address: 192.168.2.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

➢ **Network Diagram**

Laptop A
192.168.1.10
|
| Subnet: 192.168.1.0/24
|
Router R1
192.168.1.1
10.0.0.1
|
| Subnet: 10.0.0.0/30
|
Router R2
10.0.0.2

192.168.2.1
|
| Subnet: 192.168.2.0/24
|
Friend's Computer
192.168.2.20

- **Routing Table of Router R1**

Destination Network	Next Hop	Outgoing Interface
192.168.1.0/24	Directly connected	LAN
10.0.0.0/30	Directly connected	Link to R2
192.168.2.0/24	10.0.0.2	Link to R2

- **Routing Table of Router R2**

Destination Network	Next Hop	Outgoing Interface
10.0.0.0/30	Directly connected	Link to R1
192.168.2.0/24	Directly connected	Destination LAN

192.168.1.0/24	10.0.0.1	Link to R1

- **Step-by-Step Packet Journey**
- **Step 1: Packet Creation**
- The laptop creates a packet with the following information:
  - Source IP Address: 192.168.1.10
  - Destination IP Address: 192.168.2.20
- The laptop checks whether the destination belongs to its own subnet. Since 192.168.2.20 belongs to a different subnet, the laptop sends the packet to its default gateway, 192.168.1.1.

- **Step 2: Packet Reaches Router R1**
- Router R1 receives the packet and examines its destination IP address, 192.168.2.20.
- It checks its routing table and finds the entry 192.168.2.0/24, with next hop 10.0.0.2.
- Therefore, R1 forwards the packet to Router R2 through the link between the routers.
- **Step 3: Packet Reaches Router R2**
- Router R2 receives the packet and examines the destination IP address.
- Its routing table shows that 192.168.2.0/24 is directly connected to its destination LAN interface.
- Therefore, R2 forwards the packet onto the destination network.

- **Step 4: Packet Reaches the Friend's Computer**
- The friend's computer, with IP address 192.168.2.20, receives the packet.

- The destination computer can then process the data and send a reply through its default gateway, 192.168.2.1, if a response is required.

- **How the Routing Decision Is Made**
  - Routers use routing tables to select the appropriate path for packets. They generally apply the **longest prefix match** rule, choosing the most specific matching destination network.
  - For example, Router R1 finds that the destination IP 192.168.2.20 belongs to the network 192.168.2.0/24. It selects the matching route and forwards the packet to the next hop, 10.0.0.2.
  - If no specific route matches, a router may use a configured default route. If neither a matching route nor a usable default route exists, the packet may be dropped.

- **Result**
- The packet was traced conceptually from the source laptop to the friend's computer through two routers. The routing tables showed how each router selects the next hop or outgoing interface based on the destination IP address.

❖ 3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

➢ Aim

- To create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each operating on a different subnet.

➢ Introduction

- IP routing is the process of forwarding data packets from one network to another. A routing table contains information about destination networks, subnet masks, next-hop addresses, and outgoing interfaces.
- In a food delivery network such as Zomato, different city branches may use separate IP subnets. Routers use routing tables to forward communication between these branches.

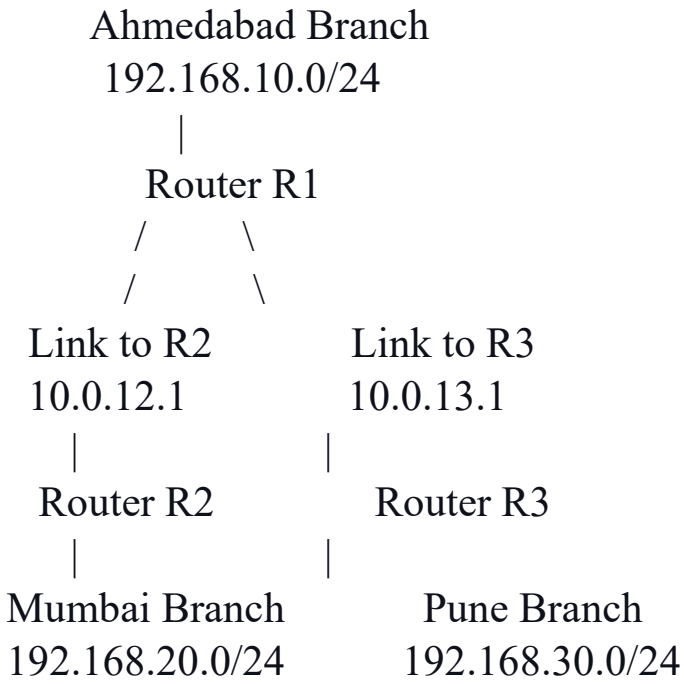
➢ Network Scenario

➢ Consider a fictional food delivery company with three branches located in different cities:

- **Ahmedabad Branch:** 192.168.10.0/24
- **Mumbai Branch:** 192.168.20.0/24
- **Pune Branch:** 192.168.30.0/24

➢ Each branch uses a different subnet. The branches communicate through a central router named **R1**, which connects to routers at the Mumbai and Pune branches.

- **Network Diagram**



- **IP Addressing Details**

Location	Network Address	Subnet Mask	Router Interface IP
Ahmedabad	192.168.10.0	255.255.255.0	192.168.10.1
Mumbai	192.168.20.0	255.255.255.0	192.168.20.1

Pune	192.168.30.0	255.255.255.0	192.168.30.1

- The router-to-router links use the following addresses:
  - **R1–R2 link:** 10.0.12.1 and 10.0.12.2
  - **R1–R3 link:** 10.0.13.1 and 10.0.13.2
- **Routing Table of Central Router R1**

Destinatin Network	Subnet Mask	Next Hop	Outgoing Interface
192.168.10.0	255.255.255.0	Directly connected	G0/0
192.168.20.0	255.255.255.0	10.0.12.2	G0/1
192.168.30.0	255.255.255.0	10.0.13.2	G0/2
10.0.12.0	255.255.255.252	Directly connected	G0/1
10.0.13.0	255.255.255.252	Directly connected	G0/2

➢ **Explanation of Routing Table Entries**

1. **Ahmedabad Network:** The network 192.168.10.0/24 is directly connected to R1, so packets for this network are sent through interface G0/0.
2. **Mumbai Network:** The network 192.168.20.0/24 is reachable through Router R2 at next hop 10.0.12.2.
3. **Pune Network:** The network 192.168.30.0/24 is reachable through Router R3 at next hop 10.0.13.2.
4. **Router Links:** The 10.0.12.0/30 and 10.0.13.0/30 networks connect R1 to the other branch routers.

➢ **Result**

➢ A routing table was prepared for a fictional food delivery network with three branches in Ahmedabad, Mumbai, and Pune. Each branch uses a different subnet, and the routing table specifies the destination network, subnet mask, next hop, and outgoing interface.

❖ **4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).**

➢ **Aim**

- To compare static routing and dynamic routing by identifying three advantages and three disadvantages of each, with examples from real-world applications and services.

- **Introduction**
  - Routing is the process of forwarding data packets from a source network to a destination network. Routers use routing tables to decide the best path for forwarding packets.
- There are two main types of routing:
  - **Static Routing:** Routes are manually configured by a network administrator.
  - **Dynamic Routing:** Routers automatically learn and update routes using routing protocols such as RIP and OSPF.
- Both methods are useful in different network environments, depending on the network size, complexity, and reliability requirements.

- **Static Routing**
- Static routing uses manually configured routes. The administrator specifies the destination network, next-hop address, or outgoing interface.
- **Advantages of Static Routing**
  - **Simple for Small Networks:** Static routing is easy to manage when only a few networks are connected. For example, a small restaurant can use a manually configured route between its office network and billing system.
  - **Low Resource Usage:** It does not require a routing protocol to exchange route information, reducing routing-related CPU and bandwidth usage.
  - **Greater Control:** The administrator decides exactly which path traffic should follow, making static routing useful for fixed network paths.

- **Disadvantages of Static Routing**

- **Manual Configuration:** Every route must be configured manually, which becomes time-consuming as the network grows.
- **No Automatic Recovery:** If a configured route fails, the administrator may need to change the route manually unless a backup route is configured.
- **Difficult to Scale:** Managing routes across many branches becomes complicated and increases the possibility of configuration errors.

➢ **Real-World Example**

➢ A small restaurant with one office and one billing network may use static routing because its network structure is simple and rarely changes.

➢ **Dynamic Routing**

➢ Dynamic routing allows routers to learn routes and update their routing tables automatically using routing protocols.

➢ Two common dynamic routing protocols are:

- **RIP (Routing Information Protocol):** Uses hop count to select routes and supports a maximum of 15 hops to a destination.
- **OSPF (Open Shortest Path First):** Uses link-state information and path cost to calculate routes. It is suitable for larger and more complex networks.

➢ **Advantages of Dynamic Routing**

1. **Automatic Route Updates:** Routers can learn new routes and update routing tables when network conditions change.
2. **Improved Adaptability:** If a network link fails, the routing protocol can recalculate routes when an alternative path is available.

3. **Suitable for Large Networks:** Protocols such as OSPF help manage routes across many interconnected networks and branches.

- **Disadvantages of Dynamic Routing**
  1. **Higher Resource Usage:** Routing protocols use CPU, memory, and some network bandwidth to exchange routing information and maintain routes.
  2. **More Complex Configuration:** Protocols such as OSPF require careful configuration and troubleshooting.
  3. **Convergence Delay:** When a network link fails, routers may need time to learn the change and calculate updated routes. Communication can be temporarily affected.

- **Real-World Example**
- A food delivery service such as Zomato may operate across multiple offices, servers, and cloud networks. Dynamic routing using OSPF within an appropriate private network can help routers adapt when a network link fails, provided an alternative path exists.
- **Important:** Dynamic routing manages network paths; it does not directly select the fastest road for a delivery rider. Rider navigation and traffic-aware delivery routes are separate systems.
- **Comparison Table**

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Routes learned automatically

Route Updates	Usually require manual changes	Updated through routing protocols
Resource Usage	Generally lower	Generally higher
Network Size	Best suited to small or stable networks	Useful for medium and large networks
Failure Handling	Requires a backup route or manual changes	Can recalculate routes when alternatives exist
Examples	Manually configured IP routes	RIP and OSPF

- **Result**
- Static routing and dynamic routing were compared by identifying three advantages and three disadvantages of each. Static routing provides simplicity and control in small networks, while dynamic routing offers automatic route learning and better adaptability in larger networks.

❖ **5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Paste the AI-generated config and annotate each part with a short description.**

➢ **Aim**

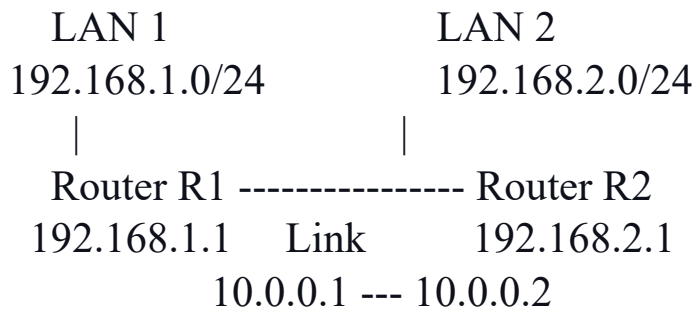
- To generate a sample OSPF configuration for two routers and explain the purpose of each configuration command.

➢ **Introduction**

- OSPF (Open Shortest Path First) is a dynamic routing protocol used to exchange routing information between routers. It automatically discovers routes and calculates the best path using a metric called **cost**.
- OSPF is commonly used in medium and large networks because it can update routing information when network conditions change.

➢ **Network Topology**

➢ Consider two routers, R1 and R2, connected through a point-to-point network.



- **IP Addressing Table**

Device	Interface	IP Address
R1	G0/0	192.168.1.1/24
R1	G0/1	10.0.0.1/30
R2	G0/0	192.168.2.1/24
R2	G0/1	10.0.0.2/30

- **AI-Generated OSPF Configuration**
- The following example uses Cisco IOS-style commands. The interface names should match the router model.
- **Configuration for Router R1**

```
enable
configure terminal
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
write memory
```

- **Explanation of R1 Commands**

Command	Explanation
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
router ospf 1	Starts an OSPF routing process with process ID 1.
router-id 1.1.1.1	Assigns a unique identifier to Router R1.
network 192.168.1.0 0.0.0.255 area 0	Enables OSPF on matching interfaces in the 192.168.1.0/24 network and advertises that network in Area 0.
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the matching router-link interface in the 10.0.0.0/30 network.
end	Exits configuration mode.
write memory	Saves the configuration.

- **Configuration for Router R2**

```
enable
configure terminal
router ospf 1
router-id 2.2.2.2
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```

end
write memory

- **Explanation of R2 Commands**

Command	Explanation
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
router ospf 1	Starts an OSPF routing process with process ID 1.
router-id 2.2.2.2	Assigns a unique identifier to Router R2.
network 192.168.2.0 0.0.0.255 area 0	Enables OSPF on matching interfaces in the 192.168.2.0/24 network and advertises that network in Area 0.
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the matching router-link interface in the 10.0.0.0/30 network.
end	Exits configuration mode.
write memory	Saves the configuration.

- **How OSPF Works**

1. R1 and R2 send OSPF Hello packets to discover neighboring routers.

2. The routers establish an OSPF neighbor relationship if their settings are compatible.
3. They exchange routing information about their connected networks.
4. Each router calculates the best available paths using the OSPF cost metric.
5. R1 learns how to reach 192.168.2.0/24 through R2, and R2 learns how to reach 192.168.1.0/24 through R1.

➢ **Verification Commands**

➢ The following commands can be used to verify OSPF operation on Cisco routers:

show ip ospf neighbor
show ip route ospf
show ip protocols

- show ip ospf neighbor displays OSPF neighbor information.
- show ip route ospf displays routes learned through OSPF.
- show ip protocols displays information about the active routing protocols.

➢ **Result**

➢ Sample OSPF configurations were prepared for two Cisco routers. The commands assign router IDs, enable OSPF for the required networks, and place the networks in Area 0. Verification commands were identified to check the neighbor relationship and learned routes.